BUSINESS INTELLIGENCE LAB

EXPERIMENT – 2

End-to-End ETL Pipeline and Sales Analysis using Microsoft Power BI

Item	Details
Course	Business Intelligence & Tools
Software	Microsoft Power BI Desktop
Dataset	Global Superstore Dataset.xlsx

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1. AIM

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

2. OBJECTIVES

- Understand the ETL process and its role in Business Intelligence.
- Extract data from the Global Superstore dataset using Microsoft Power BI.
- Clean, transform, standardize and integrate data using Power Query Editor.
- Create and validate relationships between the relevant tables in the Power BI data model.
- Create analytical measures using DAX.
- Prepare transformed data for reporting and business analysis.

3. TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor (built into Power BI Desktop)
- DAX (Data Analysis Expressions)

4. DATASET USED

Global Superstore Dataset.xlsx was used as the primary real-world business dataset. The workbook contains business information organized across worksheets such as Orders, Returns and People.

Worksheet	Role in ETL workflow
Orders	Primary sales and profitability data used for analysis.
Returns	Return information used for integration with order-level information.
People	Regional/people information used for supporting the analytical model.

5. THEORY

5.1 ETL in Business Intelligence

ETL stands for Extract, Transform and Load. It is a fundamental Business Intelligence process used to collect data from source systems, clean and modify it into a consistent structure, and load the prepared data into an analytical environment. ETL improves data usability and provides a reliable foundation for reporting and decision-making.

5.2 Extract

The Extract stage collects data from the original source without changing its business meaning. In this experiment, the Global Superstore Excel workbook was connected to Power BI and the Orders, Returns and People worksheets were selected for processing.

5.3 Transform

The Transform stage prepares extracted data for analysis. Typical operations include removing duplicate or blank records, correcting data types, renaming fields, creating derived columns, standardizing categorical values and integrating related datasets. Power Query records these operations as Applied Steps, making the transformation process transparent and repeatable.

5.4 Load

The Load stage places the transformed data into the Power BI data model. Once loaded, tables can be related and measures can be created for analysis and visualization.

5.5 Power Query Editor

Power Query Editor is Power BI's data preparation environment. It provides commands for filtering, cleaning, changing data types, merging and appending queries, adding columns and reviewing transformation steps.

5.6 Data Model and Relationships

The Power BI data model organizes tables and defines how they interact. Relationships allow filters and calculations to propagate between related tables. Order ID is the logical key for order-return integration, while Region can be used to relate Orders and People where that relationship is implemented in the model.

5.7 DAX Measures

DAX (Data Analysis Expressions) is the formula language used in Power BI for creating calculated measures. Measures calculate values dynamically according to the current filter and visualization context.

Measure	DAX expression	Purpose
Total Sales	SUM(Orders[Sales])	Overall sales value.
Total Profit	SUM(Orders[Profit])	Overall profit.
Profit Margin	DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]))	Profit as a proportion of sales.
Total Orders	DISTINCTCOUNT(Orders[Order ID])	Distinct order count.

6. PROCEDURE

The experiment was performed as an end-to-end ETL workflow: Extract → Transform → Load → Data Model → DAX Measures → Dashboard.

Part A: Extract – Connect to Dataset

Step 1 – Obtain the Global Superstore dataset

The Global Superstore Dataset.xlsx file was saved in the experiment working folder and used as the source workbook.

Purpose: To provide the original business data for extraction.

Step 2 – Connect Power BI to Excel

Power BI Desktop was opened. Home → Get Data → Excel Workbook was selected, and Global Superstore Dataset.xlsx was opened.

Purpose: To establish the connection between Power BI and the source workbook.

Step 3 – Select the required worksheets

Orders, Returns and People were selected from the workbook. Transform Data was used so the tables could be prepared in Power Query before loading.

Purpose: To bring the required source tables into the ETL pipeline.

Part B: Transform – Power Query Editor

Step 4 – Open and inspect Power Query Editor

The Queries pane was checked to verify that Orders, Returns and People were available for transformation.

Purpose: To provide a controlled environment for data cleaning and transformation.

Step 5 – Transform the Orders table

Duplicate records and blank rows were handled where applicable. Data types were verified, required column names were standardized, Year and Month were derived from Order Date, a Profit Margin field was prepared, and Region and Segment values were standardized.

Purpose: To create a clean and analysis-ready primary sales table.

Step 6 – Transform the Returns table

The Returns table was inspected for duplicate Order IDs and missing values. Returned field values were checked and standardized.

Purpose: To make return information consistent before integration.

Step 7 – Transform the People table

Regional assignments and region-name consistency were checked. Unnecessary columns were removed where required and the remaining fields were verified.

Purpose: To prepare supporting regional/people information for the model.

Step 8 – Merge Orders and Returns

Orders and Returns were integrated using Order ID as the matching field through Merge Queries and the appropriate join condition.

Purpose: To combine order information with return information at the order level.

Step 9 – Review Applied Steps and load

All Applied Steps were reviewed. Home → Close & Apply was then used to load the prepared data into Power BI.

Purpose: To complete the Transform stage and load the cleansed data.

Part C: Load – Build Data Model

Step 10 – Open Model View

Model View was opened from the left navigation pane to inspect the loaded analytical model.

Purpose: To verify the organization of the loaded tables.

Step 11 – Create or verify relationships

Relationships were created or verified using the relevant business keys. Order ID was used for order-return integration, and Region was used for Orders-People where applicable to the implemented model.

Purpose: To allow related information to work together during analysis.

Step 12 – Create DAX measures

New Measure was used in the Orders table to create Total Sales, Total Profit, Profit Margin and Total Orders. These measures were used in KPI cards and charts.

Purpose: To generate reusable business performance indicators.

Step 13 – Build the BI dashboard

A report page was designed with KPI cards for Total Sales, Total Profit, Profit Margin and Total Orders; category-wise sales and profit visuals; a Profit Margin vs Discount scatter chart; and a detailed sales/profit table. The title was set to GLOBAL SUPERSTORE – ETL & SALES ANALYSIS.

Purpose: To convert the prepared data into a clear business intelligence report.

7. RESULT AND SCREENSHOT EVIDENCE

The ETL pipeline was successfully implemented in Microsoft Power BI. The Global Superstore data was prepared for analysis and a consolidated sales and profitability dashboard was created. The final Power BI output captured from the practical is shown below.

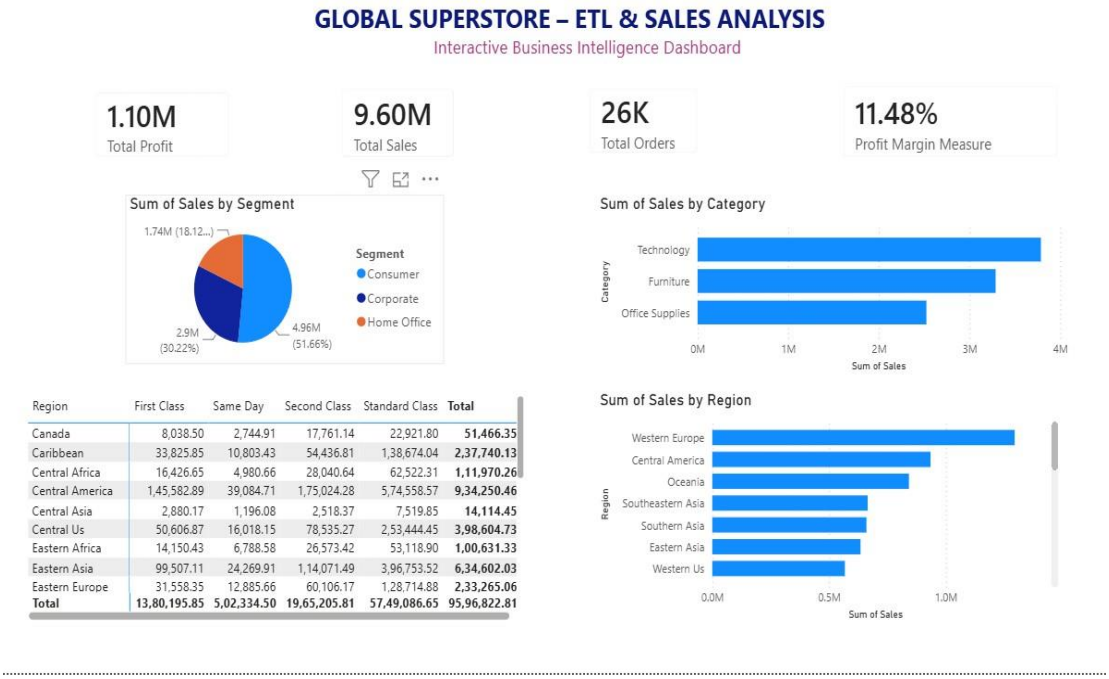


Figure 1: Final Power BI dashboard – GLOBAL SUPERSTORE – ETL & SALES ANALYSIS.

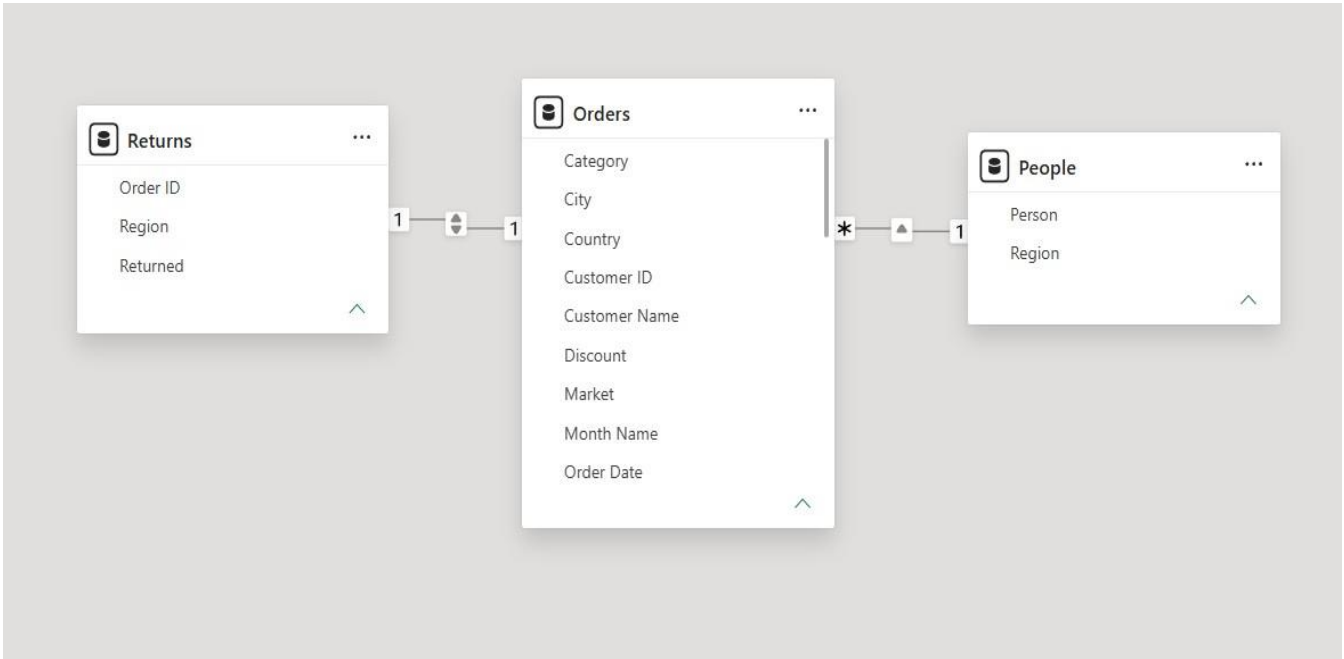


Figure 2 : Power BI Model View showing relationships between Orders, Returns, and People tables.

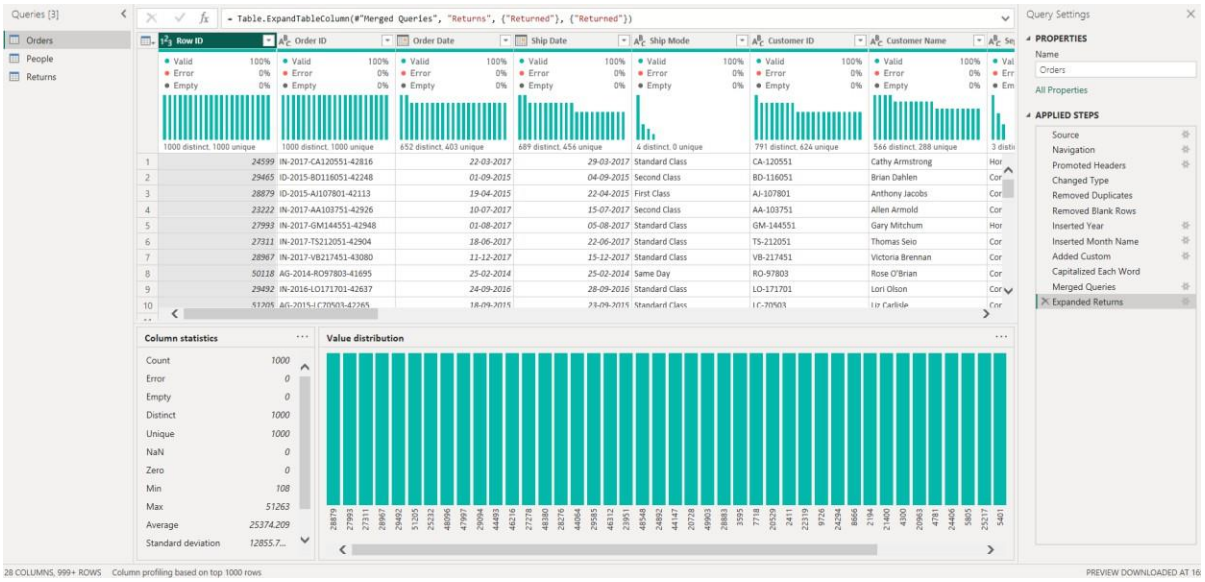


Figure 3: Data transformation and cleaning of the Orders table in Power Query Editor.

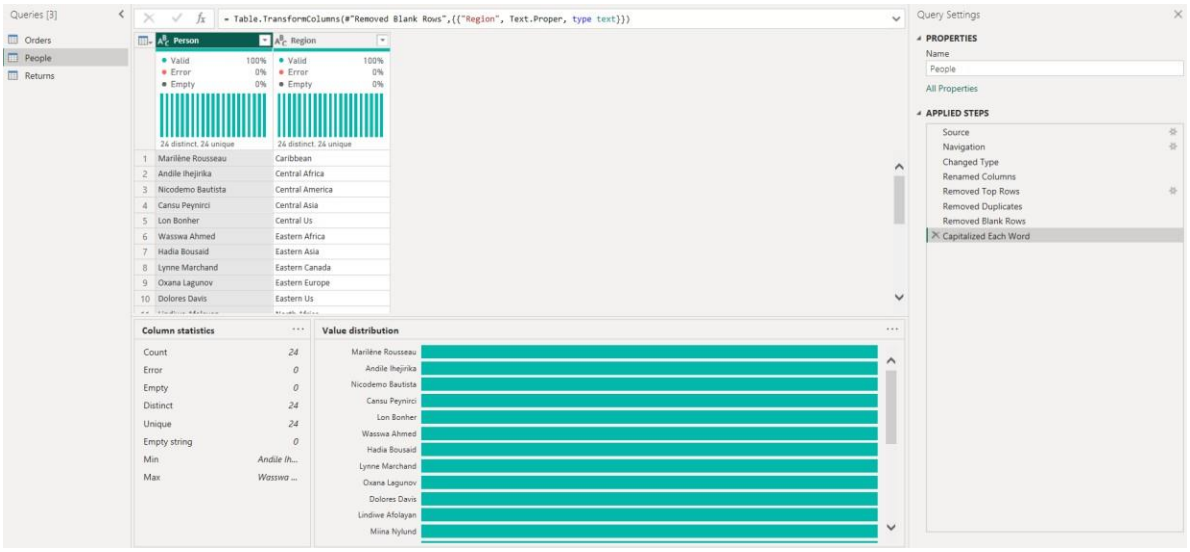


Figure 4: Data transformation and validation of the People table in Power Query Editor.

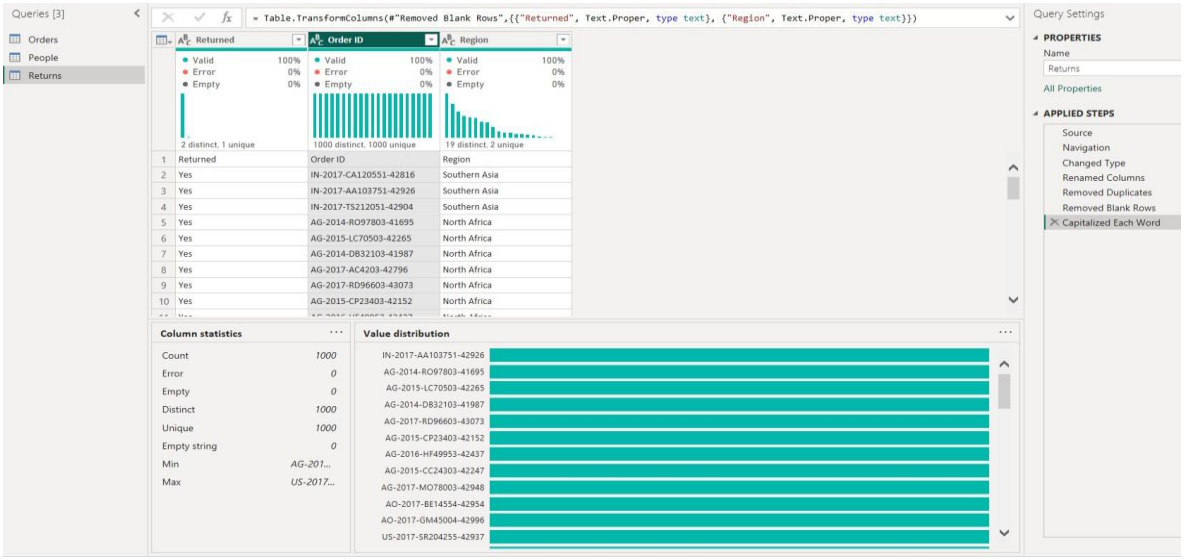


Figure 5: Data transformation and validation of the Returns table in Power Query Editor.

7.1 Dashboard Components

Visual / KPI	What it represents
Sales by Category	Compares total sales across Technology, Furniture and Office Supplies.
Profit by Category	Shows the contribution of each category to total profit.
Total Sales	Displays the overall sales value using the Total Sales measure.
Total Profit	Displays the overall profit using the Total Profit measure.
Profit Margin vs Discount	Shows the relationship between discount and average profit margin by category.
Profit Margin	Displays the calculated profit margin KPI.
Total Orders	Displays the distinct order count.
Detailed Sales & Profit Analysis	Provides category, city, country, state, sales and profit information.

7.2 Observed Dashboard Output

The captured dashboard displays approximately 12.64M Total Sales, 1.47M Total Profit, a Profit Margin of approximately 0.12 and approximately 26K Total Orders. The category visuals compare Technology, Furniture and Office Supplies, while the detailed table provides city-, country- and state-level sales and profit information.

8. DATA TRANSFORMATION AND ANALYTICAL OBSERVATIONS

- The source workbook was converted into an analytical Power BI workflow through Power Query.
- Cleaning and standardization were performed before reporting so that the data could be used consistently.
- Order-level and supporting information were prepared for integration through business keys such as Order ID and Region.
- DAX measures converted raw sales and profit fields into reusable business KPIs.
- The final report combines high-level KPIs with category-level and detailed analytical views.
- The dashboard provides a consolidated view of sales performance, profitability, order volume and discount-related behaviour.

9. LEARNING OUTCOMES

- Understood the ETL process and its role in Business Intelligence.
- Learned how to extract data from a real-world Excel business dataset using Power BI.
- Gained practical experience with Power Query Editor.
- Learned how to clean, filter, transform and integrate datasets.
- Learned how to handle duplicate records, blank values and data-type inconsistencies.
- Understood the purpose of relationships in the Power BI data model.
- Learned how to create analytical measures using DAX.
- Understood the use of Total Sales, Total Profit, Profit Margin and Total Orders.
- Learned how to create KPI cards, charts and detailed tables for BI reporting.

10. CONCLUSION

The experiment successfully demonstrated an end-to-end ETL workflow in Microsoft Power BI. The Global Superstore dataset was extracted from Excel, transformed using Power Query Editor, integrated into an analytical model and used to create DAX-based business measures. The final dashboard presents sales, profit, margin, orders and category-level analysis in a unified report. This workflow demonstrates how raw business data can be converted into structured information that supports reporting and informed decision-making.